

Yash Kherwal

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careerautomated.in

EDUCATION

Indian Institute of Technology Roorkee
B.Tech in Chemical Engineering

2022 – 2026

EXPERIENCE

AI Engineer Intern

Feb 2026 – Apr 2026

Orange Labs — EdTech ERP platform for schools and colleges

- Developed computer vision model detecting students from live camera feeds, auto-syncing presence records into the school ERP, eliminating manual marking.
- Engineered synthesised audio/video from real teacher input, improving engagement over generic video content.

Software Development Intern

May 2025 – June 2025

ScoreMe Solutions

- Designed rule-based heuristics for fintech PDF processing, handling deterministic inputs first, with Llm resolving ambiguous cases.
- Implemented **confidence calibration layer**: predictions below threshold withheld from **credit scoring pipelines** and flagged for review; integrated **Tesseract OCR** as fallback for scanned inputs.

Engineering Intern

June 2025 – July 2025

Bharat Electronics Limited

- Implemented a **binary stream parser** for **ASTERIX CAT048** radar protocol per EUROCONTROL spec; designed **concurrent ingestion** with **backpressure handling** to maintain throughput under sustained high-frequency surveillance loads.

PROJECTS

CareerAutomated — Autonomous AI Recruiting Platform | *Python, Playwright, LangGraph, RAG*

2025 – 2026

- Shipped an **LLM-based resume tailoring engine** under an **integrity gate** that fact-checks every rewritten claim against the source resume, blocking hallucinated content before it ships.
- Developed an end-to-end AI job platform: **50+ ATS discovery connectors** ingest and normalise **1.3M+ live job postings**; a **hybrid BM25 + semantic RAG** pipeline scores per-candidate job fit against real profile data.
- Engineered a **plugin-based auto-apply engine** — a shared Playwright orchestration core with per-ATS handlers (Greenhouse/Lever/Ashby) fills and submits real applications; **human-in-the-loop safety gates** block submission rather than guess when required data is missing.

Hybrid RAG Retrieval System | *Python, LangChain, BGE-M3, AstraDB, FastAPI, AWS EC2*

2024

- Diagnosed keyword retrieval failure on 500+ document corpus — BM25 missed semantically relevant results due to vocabulary mismatch; introduced **hybrid retrieval** combining **BGE-M3** dense embeddings with sparse BM25 scores, improving recall on paraphrased and domain-specific queries.
- Designed **decoupled ingestion and serving** on **AWS EC2** for zero-downtime corpus updates; **response groundedness** enforced via prompt constraints to prevent hallucination.

YAAR — Personality-Adaptive LLM System | *React Native, LLM APIs, FastAPI, Persistent Memory*

2025

- Shipped production **Android APK** via React Native + EAS Build — validated end-to-end on device with sub-2s streamed LLM response latency.
- Designed a **persistent personality memory system**: each interaction updates a **personality metric vector** (humour tolerance, self-awareness index, reaction intensity) injected into the **LLM context window** at inference time to steer tone without fine-tuning.

TECHNICAL SKILLS

LLM & Agentic AI: LangGraph, LangChain, CrewAI, Multi-Agent Orchestration, MCP, Structured Outputs, Prompt Engineering, Agentic Workflows

Retrieval: Hybrid RAG, BGE-M3, Dense + Sparse Retrieval, AstraDB, Vector Databases, ANN Indexing

ML / AI: TensorFlow, Scikit-learn, Random Forest, Confidence Calibration, Computer Vision (OpenCV)

Backend & Infra: FastAPI, Docker, AWS EC2, n8n, REST APIs, Stream Processing

Languages & Data: Python, SQL, Java, C++, DSA, Pandas, NumPy